UNIVERSIDAD CARLOS III DE MADRID
Applied Mathematics and Computation
Applied Functional Analysis
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Mini-Project: Boundedness and Extensions of Differential Operators in Hilbert Spaces

In many applications of the theory of operators on Hilbert spaces the operators we have to deal with are differential operators, that is operators involving functions and their derivatives. In this mini-project we will try to get familiar with some basic properties of differential operators and some ideas to work with them, like the introduction of Sobolev spaces of functions.

Your final report should clearly explain your reasoning, include any computations or plots, and indicate when you used AI assistance.

Part 1: The Operator $T = \frac{d}{dx}$ on $L^2(\mathbb{R})$

Recall that

$$L^2(\mathbb{R}) = \{f : \mathbb{R} \to \mathbb{C} \mid \|f\|_{L^2}^2 = \int_{\mathbb{R}} |f(x)|^2 \, dx < \infty\}.$$

Consider the differentiation operator

$$(Tf)(x) = f'(x),$$

with domain

$$D(T) = \{f \in L^2(\mathbb{R}) \mid f' \in L^2(\mathbb{R})\}.$$

Task 1. The operator T is not bounded.

- Explain what it means for an operator $T : \mathcal{H} \to \mathcal{H}'$ between Hilbert spaces to be *bounded*.
- Show that T is **not bounded** as an operator on $L^2(\mathbb{R})$. To do this, construct a sequence (f_n) in $L^2(\mathbb{R})$ such that

$$\|f_n\|_{L^2} = 1 \quad \text{but} \quad \|Tf_n\|_{L^2} \to \infty.$$

- Verify your conclusion numerically: plot $\|Tf_n\|_{L^2}$ versus n using a discrete approximation of the derivative.

Task 2. The Sobolev Space $H^1(\mathbb{R})$ and Boundedness

- Define the Sobolev space

$$H^1(\mathbb{R}) = \{f \in L^2(\mathbb{R}) \mid f' \in L^2(\mathbb{R})\},$$

  with inner product

$$\langle f, g\rangle_{1,2} = \int_{-\infty}^{+\infty} \overline{f(x)}g(x) + \overline{f'(x)}g'(x)\, dx\,.$$

  Show that $\langle f, g\rangle_{1,2}$ is an inner product.

- Study the properties of the norm $|| \cdot ||_{1,2}$ associated to the inner product $\langle f, g\rangle_{1,2}$.
- Can you check the completeness of the space $H^1(\mathbb{R})$?
- Show that $T : H^1(\mathbb{R}) \to L^2(\mathbb{R})$ is bounded. Compute or estimate its operator norm:

$$\|T\| = \sup_{f\neq 0} \frac{\|Tf\|_{L^2}}{\|f\|_{H^1}}.$$

- Compute the adjoint $T^\dagger$ of T.
- Use your AI assistant to explain, in your own words, why adding information about the derivative into the norm (as in H^1) "controls" the growth of $\|Tf\|$.

Part 2: Other differential operators and boundary conditions

Task 3. Differential operators on Hilbert spaces over compact intervals.

- Study now the operator $T = \frac{d}{dx}$ on the Hilbert space $L^2([0,1])$ with different domains defined by boundary conditions:
  - Dirichlet boundary conditions: The domain of the operator is given by $D(T)_{\text{Dirichlet}} = \{f \in L^2([0,1]) \mid f' \in L^2([0,1]), f(0) = f(1) = 0\}$;
  - Neumann boundary conditions: $D(T)_{\text{Neumann}} = \{f \in L^2([0,1]) \mid f' \in L^2([0,1]), f'(0) = f'(1) = 0\}$;
  - Periodic boundary conditions: $D(T)_{\text{Periodic}} = \{f \in L^2([0,1]) \mid f' \in L^2([0,1]), f(0) = f(1), f'(0) = f'(1)\}$.

    Compare the domains, boundedness, and the form of the adjoint operator in each case.

Task 4. Study the Laplacian operator.

Consider the Laplacian operator $\Delta = \frac{d^2}{dx^2}$ on $L^2(\mathbb{R})$ or $L^2([0,1])$, and study its continuity. Introduce the Sobolev space H^2 and discuss boundedness of $\Delta : H^2 \to L^2$.

Task 5. Integral operators.

Explore the integral operator

$$(If)(x) = \int_0^x f(t)\,dt,$$

and show that it *is* bounded on $L^2([0,1])$. Estimate its operator norm.

Part 3: Optional Numerical Exploration

Task 6. Using `Python`, `NumPy`, or a similar tool:

- Discretize functions on a uniform grid in $[0,1]$.
- Represent differentiation by a finite difference matrix.
- Compute its matrix norm and observe how it behaves as the grid spacing decreases.

Explain how this numerical behavior reflects the theoretical unboundedness of T on L^2.

Deliverables

- A short written report (4-6 pages) including definitions, reasoning, and any plots or computations.
- A short note describing how you used the AI assistant during the project.

Grading criteria and notes

Conceptual understanding and explanations	40%
Mathematical correctness and computations	30%
Creativity and depth of exploration	20%
Clarity and presentation	10%

Use your AI assistant as a mathematical collaborator: ask it to check derivations, clarify definitions, suggest examples, or produce Python snippets. However, always ensure that the reasoning and final presentation are your own.